

Qualitative and Limited Dependent Variable Models

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1 Introduction

In this chapter, we will:

- Examine models that are used to describe choice behavior, and which do not have the usual continuous dependent variable
- Introduce a class of models with dependent variables that are limited
 - They are continuous, but that their range of values is constrained in some way, and their values not completely observable
 - Alternatives to least squares estimation are needed since the least squares estimator is biased and inconsistent

2 Models with Binary Dependent Variables

- Many of the choices that individuals and firms make are “either–or” in nature
- Such choices can be represented by a binary (indicator) variable that takes the value 1 if one outcome is chosen and the value 0 otherwise

- The binary variable describing a choice is the *dependent variable* rather than an independent variable
- **Examples:**
 - Models of why some individuals take a second or third job, and engage in “moonlighting”
 - Models of why some legislators in the U.S. House of Representatives vote for a particular bill and others do not
 - Models explaining why some loan applications are accepted and others are not at a large metropolitan bank
 - Models explaining why some individuals vote for increased spending in a school board election and others vote against
 - Models explaining why some female college students decide to study engineering and others do not

Example: Driving to work vs. taking the bus to work?

- We represent an individual’s choice by the indicator variable:

$$y = \begin{cases} 1 & \text{If individual drives to work} \\ 0 & \text{If individual takes bus to work} \end{cases} \quad (1)$$

- If the probability that an individual drives to work is p , then $\Pr(y = 1) = p$
- The probability that a person uses public transportation is $\Pr(y = 0) = 1 - p$
- The probability distribution function for such a binary random variable is:

$$f(y) = p^y (1 - p)^{1-y}, \quad y = 0, 1 \quad (2)$$

with

$$\begin{aligned} E(y) &= p \\ \text{Var}(y) &= p(1 - p) \end{aligned}$$

- As an example, consider a transportation problem of having to choose between driving to work or taking the bus. For our analysis, can define the explanatory variable as:

$$x = (\text{commuting time by bus} - \text{commuting time by car})$$

- We could model the indicator variable y using the linear model, however, there are several problems:
 - It implies marginal effects of changes in continuous explanatory variables are constant, which cannot be the case for a probability model
 - This feature also can result in predicted probabilities outside the $[0, 1]$ interval
 - The linear probability model error term is heteroskedastic, so that a better estimator is generalized least squares
- In regression analysis we break the dependent variable into fixed and random parts
- If we do this for the indicator variable y , we have:

$$y = E(y) + e = p + e \quad (3)$$

- Assuming that the relationship is linear:

$$E(y) = p = \beta_1 + \beta_2 x \quad (4)$$

- The linear regression model for explaining the choice variable y is called the *linear probability model*:

$$y = E(y) + e = p = \beta_1 + \beta_2 x + e \quad (5)$$

- Using these values it can be shown that the variance of the error term e is:

$$Var(e) = (\beta_1 + \beta_2 x) (1 - \beta_1 - \beta_2 x)$$

- The estimated variance of the error term is:

$$\sigma_i^2 = \widehat{Var}(e_i) = (b_1 + b_2 x_i) (1 - b_1 - b_2 x_i) \quad (6)$$

- We can transform the data as:

$$\begin{aligned} y_i^* &= \frac{y_i}{\widehat{\sigma}_i} \\ x_i^* &= \frac{x_i}{\widehat{\sigma}_i} \end{aligned}$$

- Then estimate the model:

$$y_i^* = \beta_1 \frac{1}{\widehat{\sigma}_i} + \beta_2 x_i^* + e_i^*$$

by least squares to produce the *feasible generalized least squares* estimates

- Both least squares and feasible generalized least squares are consistent estimators of the regression parameters
- If we estimate the parameters of Eq. 5 by least squares, we obtain the fitted model explaining the systematic portion of y
- This systematic portion is p

$$\widehat{p} = b_1 + b_2 x \quad (7)$$

- By substituting alternative values of x , we can easily obtain values that are less than zero or greater than one
- The underlying feature that causes these problems is that the linear probability model implicitly assumes that increases in x have a constant effect on the probability of choosing to drive

$$\frac{dp}{dx} = \beta_2 \quad (8)$$

- To keep the choice probability p within the interval $[0, 1]$, a nonlinear S-shaped relationship between x and p can be used

- A functional relationship that is used to represent such a curve is the probit function
- The probit function is related to the standard normal probability distribution:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

- The probit function is:

$$\Phi(z) = \Pr(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \quad (9)$$

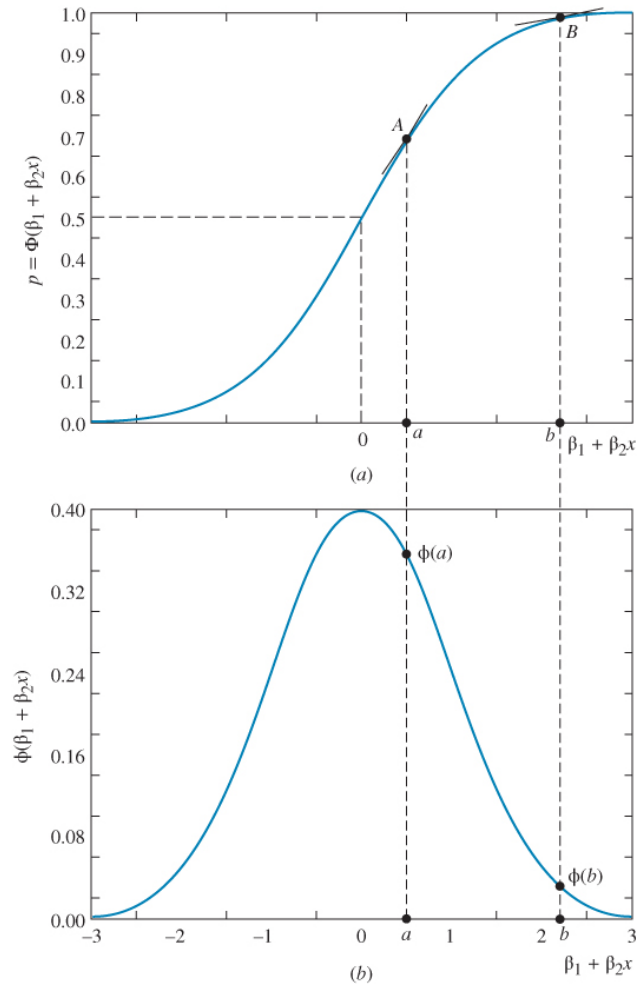


Figure 1: (a) Standard normal cumulative distribution function (b) Standard normal probability density function

- The probit statistical model expresses the probability p that y takes the value 1 to be:

$$p = \Pr(Z \leq \beta_1 + \beta_2 x) = \Phi(\beta_1 + \beta_2 x) \quad (10)$$

- The probit model is said to be nonlinear
- We can examine the *marginal effect* of a one-unit change in x on the probability that $y = 1$ by considering the derivative:

$$\frac{dp}{dx} = \frac{d\Phi(t)}{dt} \frac{dt}{dx} = \phi(\beta_1 + \beta_2 x) \beta_2 \quad (11)$$

- Eq. 11 has the following implications:

1. Since $\Phi(\beta_1 + \beta_2 x)$ is a probability density function, its value is always positive
2. As x changes, the value of the function $\Phi(\beta_1 + \beta_2 x)$ changes
3. If $\beta_1 + \beta_2 x$ is large, then the probability that the individual chooses to drive is very large and close to one

- We estimate the probability p to be:

$$\hat{p} = \Phi(\hat{\beta}_1 + \hat{\beta}_2 x) \quad (12)$$

- By comparing to a threshold value, like 0.5, we can predict choice using the rule:

$$\hat{y} = \begin{cases} 1 & \text{if } \hat{p} \geq 0.5 \\ 0 & \text{if } \hat{p} < 0.5 \end{cases}$$

- The probability function for y is combined with the probit model to obtain:

$$f(y_i) = [\Phi(\beta_1 + \beta_2 x)]^{y_i} [1 - \Phi(\beta_1 + \beta_2 x)]^{1-y_i}, \quad y_i = 0, 1 \quad (13)$$

Example: (continued)

- If the three individuals are independently drawn, then:

$$f(y_1, y_2, y_3) = f(y_1) f(y_2) f(y_3)$$

- The probability of observing $y_1 = 1, y_2 = 1$, and $y_3 = 0$ is:

$$\Pr(y_1 = 1, y_2 = 1, y_3 = 0) = f(1, 1, 0) = f(1) f(1) f(0)$$

- We now have:

$$\begin{aligned} \Pr(y_1 = 1, y_2 = 1, y_3 = 0) \\ &= \Phi(\beta_1 + \beta_2 \times 15) \Phi(\beta_1 + \beta_2 \times 6) [1 - \Phi(\beta_1 + \beta_2 \times 7)] \\ &= L(\beta_1, \beta_2) \end{aligned} \quad (14)$$

for $x_1 = 15, x_2 = 6$, and $x_3 = 7$

- This function, which gives us the probability of observing the sample data, is called the *likelihood function*
- The notation $L(\beta_1, \beta_2)$ indicates that the likelihood function is a function of the unknown parameters, β_1 and β_2

- In practice, instead of maximizing Eq. 14, we maximize the logarithm of Eq. 14, which is called the log-likelihood function:

$$\begin{aligned}\ln L(\beta_1, \beta_2) &= \{\ln \Phi(\beta_1 + \beta_2 \times 15) \Phi(\beta_1 + \beta_2 \times 6) \\ &\quad \times [1 - \Phi(\beta_1 + \beta_2 \times 7)]\} \\ &= \ln \Phi(\beta_1 + \beta_2 \times 15) + \ln \Phi(\beta_1 + \beta_2 \times 6) \\ &\quad + \ln [1 - \Phi(\beta_1 + \beta_2 \times 7)]\end{aligned}\tag{15}$$

- The maximization of the log-likelihood function is easier than the maximization of Eq. 14
- The values that maximize the log-likelihood function also maximize the likelihood function
- These are the *maximum likelihood estimates*
- A feature of the maximum likelihood estimation procedure is that while its properties in small samples are not known, we can show that in large samples the maximum likelihood estimator is normally distributed, consistent and best, in the sense that no competing estimator has smaller variance

Example: Driving to work vs. taking the bus to work?

- Let

$$DTIME = \frac{(BUSTIME - AUTOTIME)}{10},$$

which is the commuting time differential in 10-minute increments

- The probit model is:

$$P(AUTO = 1) = \Phi(\beta_1 + \beta_2 DTIME)$$

- The maximum likelihood estimates of the parameters are:

$$\begin{array}{ccc}\hat{\beta}_1 + \hat{\beta}_2 DTIME &= & -0.0644 \quad 0.3000 DTIME_i \\ (se) & & (0.3992) \quad (0.1029)\end{array}$$

- The marginal effect of increasing public transportation time, given that travel via public transportation currently takes 20 minutes longer than auto travel is:

$$\begin{aligned}\frac{\widehat{dp}}{dDTIME} &= \phi(\hat{\beta}_1 + \hat{\beta}_2 DTIME) \times \hat{\beta}_2 \\ &= \phi(-0.0644 + 0.3000 \times 2) \times 0.3000 \\ &= 0.1037\end{aligned}$$

- If an individual is faced with the situation that it takes 30 minutes longer to take public transportation than to drive to work, then the estimated probability that auto transportation will be selected is:

$$\begin{aligned}\hat{p} &= \Phi(\hat{\beta}_1 + \hat{\beta}_2 DTIME) \\ &= \Phi(-0.0644 + 0.3000 \times 3) \\ &= 0.7983\end{aligned}$$

- Since $0.7983 > 0.5$, we “predict” the individual will choose to drive
- Rather than evaluate the marginal effect at a specific value, or the mean value, the *average marginal effect* (AME) is often considered:

$$\widehat{AME} = \frac{1}{N} \sum_{i=1}^N \phi(\hat{\beta}_1 + \hat{\beta}_2 DTIME_i) \times \hat{\beta}_2$$

- For our problem: $\widehat{AME} = 0.0484$
- The sample standard deviation is: 0.0365
- Its minimum and maximum values are 0.0025 and 0.1153
- Consider the marginal effect:

$$\frac{\widehat{dp}}{dDTIME} = \phi \left(\widehat{\beta}_1 + \widehat{\beta}_2 DTIME \right) \times \widehat{\beta}_2 = g \left(\widehat{\beta}_1, \widehat{\beta}_2 \right)$$

- The marginal effect function is *nonlinear*

R Example

```
rm(list=ls())
library(POE5Rdata)

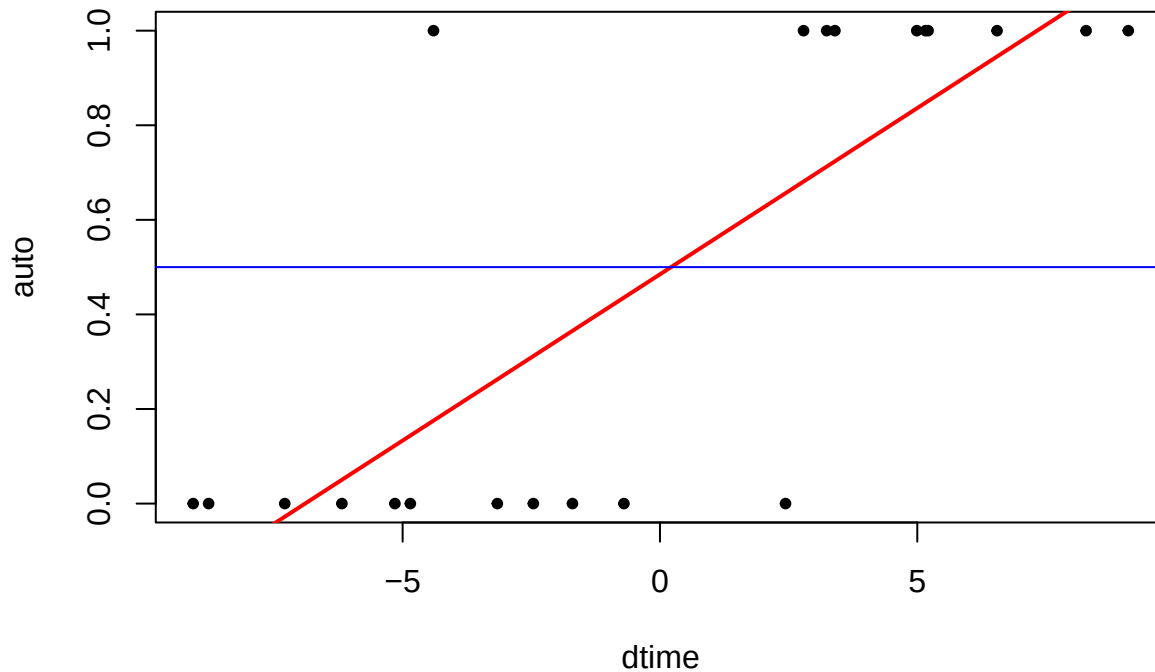
# Read in the data
data("transport", package="POE5Rdata")
attach(transport)

# Mode 1: OLS Model
prob.mod = lm(auto ~ dtime, data=transport)

summary(prob.mod)

##
## Call:
## lm(formula = auto ~ dtime, data = transport)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.65635 -0.14379  0.02776  0.15292  0.82457
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.48480     0.07145   6.785 1.76e-06 ***
## dtime         0.07031     0.01286   5.467 2.83e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3273 on 19 degrees of freedom
## Multiple R-squared:  0.6113, Adjusted R-squared:  0.5909
## F-statistic: 29.88 on 1 and 19 DF,  p-value: 2.834e-05

plot(dtime, auto, pch=20)
abline(prob.mod, col="red", lwd=2)
abline(h=0.5, col="blue")
```



```
# y_hat < 0
prob.mod$fitted[prob.mod$fitted<0]

##           8           11           20
## -0.13182290 -0.15291587 -0.02776422

# y_hat > 1
prob.mod$fitted[prob.mod$fitted>1]

##           3           6
## 1.066961 1.124615

# Predict probabilities:
# Case 1: bus takes 30 min longer than car
xdtime <- data.frame(dtime=3) # 30 min difference
predict(prob.mod,xdtime)

##           1
## 0.6957248

# Case 2: bus takes 30 min less than car
xdtime <- data.frame(dtime=-3) # 30 min difference
predict(prob.mod,xdtime)

##           1
## 0.2738653

# Model 2: Probit
probit.mod = glm(auto~dtime, family=binomial(link="probit"),
                 data=transport)
summary(probit.mod)

##
## Call:
## glm(formula = auto ~ dtime, family = binomial(link = "probit"),
##      data = transport)
```

```

##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.65981  -0.36468  -0.07317   0.37806   2.23046
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.06443    0.40068  -0.161  0.87224
## dttime       0.29999    0.10290   2.915  0.00355 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 29.065  on 20  degrees of freedom
## Residual deviance: 12.330  on 19  degrees of freedom
## AIC: 16.33
##
## Number of Fisher Scoring iterations: 7

```

```

confint(probit.mod)

##              2.5 %    97.5 %
## (Intercept) -0.8686690 0.7449619
## dttime       0.1331644 0.5435953

```

```

# Average Marginal Effects
sum_phi <- mean(dnorm(predict(probit.mod,, type = "link")))
ame = sum_phi*coef(probit.mod)
print(ame)

## (Intercept)      dttime
## -0.01039727  0.04840695

```

```

# Predict probabilities:
# Case 1: bus takes 30 min longer than car
xdtime <- data.frame(dttime=3) # 30 min difference
predict(probit.mod, xdtime, type = "response", se.fit = TRUE)

## $fit
##      1
## 0.7982915
##
## $se.fit
##      1
## 0.1365455
##
## $residual.scale
## [1] 1

```

```

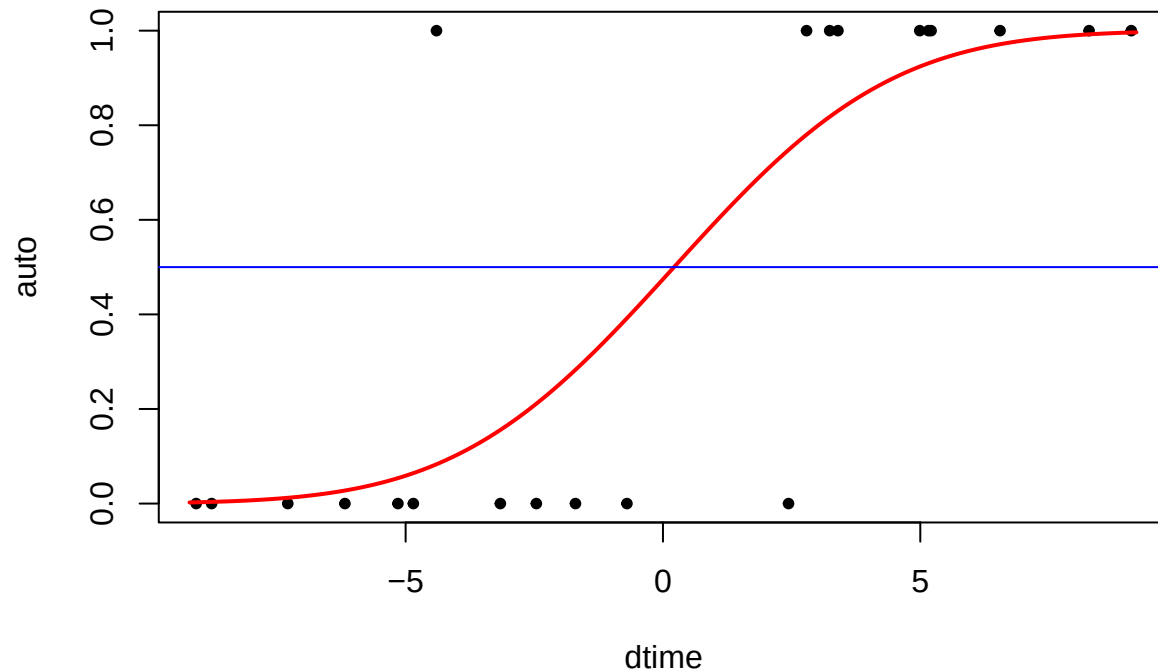
# Case 2: bus takes 30 min less than car
xdtime <- data.frame(dttime=-3) # 30 min difference
predict(probit.mod, xdtime, type = "response", se.fit = TRUE)

## $fit
##      1
## 0.167422

```

```
##
## $se.fit
##      1
## 0.1316944
##
## $residual.scale
## [1] 1

x = seq(-9.2, 9.2, 0.01)
yhat = predict(probit.mod, list(dtime = x), type = "response", se.fit = TRUE)
plot(dtime, auto, pch=20)
lines(x, yhat$fit, lwd=2, col="red")
abline(h=0.5, col="blue")
```



3 The Logit Model for Binary Choice

- Probit model estimation is numerically complicated because it is based on the normal distribution
- A frequently used alternative to the probit model for binary choice situations is the logit model
- These models differ only in the particular S-shaped curve used to constrain probabilities to the $[0, 1]$ interval
- If L is a logistic random variable, then its probability density function is:

$$\lambda(l) = \frac{e^{-l}}{(1 + e^{-l})^2}, \quad \text{for } -\infty \leq l \leq \infty \quad (16)$$

- The cumulative distribution function for a logistic random variable is:

$$\Lambda(l) = \Pr(L \leq l) = \frac{1}{1 + e^{-l}} \quad (17)$$

- The probability p that the observed value y takes the value 1 is:

$$\Pr(L \leq \beta_1 + \beta_2 x) = \Lambda(\beta_1 + \beta_2 x) = \frac{1}{1 + e^{-(\beta_1 + \beta_2 x)}} \quad (18)$$

- The probability that $y = 1$ is:

$$p = \frac{1}{1 + e^{-(\beta_1 + \beta_2 x)}} = \frac{\exp(\beta_1 + \beta_2 x)}{1 + \exp(\beta_1 + \beta_2 x)}$$

- The probability that $y = 0$ is:

$$1 - p = \frac{1}{1 + \exp(\beta_1 + \beta_2 x)}$$

- The shapes of the logistic and normal probability density functions are somewhat different, and maximum likelihood estimates of β_1 and β_2 will be different
- However, the marginal probabilities and the predicted probabilities differ very little in most cases

Example: Coke vs. Pepsi.

$$COKE = \begin{cases} 1 & \text{If Coke is chosen} \\ 0 & \text{If Pepsi is chosen} \end{cases}$$

- Based on “scanner” data on 1,140 individuals who purchased Coke or Pepsi, the probit and logit models for the choice are:

$$\begin{aligned} \text{Probit : } p_{\text{Coke}} &= E(COKE) \\ &= \Phi(\beta_1 + \beta_2 PRATIO + \beta_3 DISP_COKE + \beta_4 DISP_PEPSI) \\ \text{Logit : } p_{\text{Coke}} &= E(COKE) \\ &= \Lambda(\gamma_1 + \gamma_2 PRATIO + \gamma_3 DISP_COKE + \gamma_4 DISP_PEPSI) \end{aligned}$$

	LPM	probit	logit
<i>C</i>	0.89022*** (0.0653)	1.10806*** (0.1900)	1.92297*** (0.3258)
<i>PRATIO</i>	-0.40086*** (0.0604)	-1.14596*** (0.1809)	-1.99574*** (0.3146)
<i>DISP_COKE</i>	0.07717** (0.0339)	0.21719** (0.0966)	0.35160** (0.1585)
<i>DISP_PEPSI</i>	-0.16566*** (0.0344)	-0.44730*** (0.1014)	-0.73099*** (0.1678)

Standard errors in parentheses (White robust se for LPM) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 2: Coke-Pepsi Models

- The parameters and their estimates vary across the models and no direct comparison is very useful, but some rules of thumb exist
- Roughly:

$$\begin{aligned} \hat{\gamma}_{\text{Logit}} &\simeq 4\hat{\beta}_{\text{LPM}} \\ \hat{\beta}_{\text{Probit}} &\simeq 2.5\hat{\beta}_{\text{LPM}} \\ \hat{\gamma}_{\text{Logit}} &\simeq 1.6\hat{\beta}_{\text{Probit}} \end{aligned}$$

3.1 Wald Hypothesis Tests

- If the null hypothesis is $H_0: \beta_k = c$, then the test statistic using the probit model is:

$$t = \frac{\hat{\beta}_k - c}{se(\hat{\beta}_k)} \sim t_{(N-K)}$$

- The t -test is based on the *Wald principle*
- Using the probit model, consider the two hypotheses:

- Hypothesis 1:

$$H_0: \beta_3 = -\beta_4, \quad H_1: \beta_3 \neq -\beta_4$$

- Hypothesis 2:

$$H_0: \beta_3 = \beta_4 = 0, \quad H_1: \beta_3 \text{ and/or } \beta_4 \text{ is not } 0$$

- To test hypothesis (1) in a linear model, we would compute:

$$t = \frac{\hat{\beta}_{DISP_COKE} + \hat{\beta}_{DISP_PEPSI}}{se(\hat{\beta}_{DISP_COKE} + \hat{\beta}_{DISP_PEPSI})} \sim t_{(1,140-4)}$$

- Noting that it is a two-tail hypothesis, we reject the null hypothesis at the $\alpha = 0.05$ level if $t \geq 1.96$ or $t \leq -1.96$
- The calculated t -value is $t = -2.3247$, so we reject the null hypothesis
- We conclude that the effects of the Coke and Pepsi displays are not of equal magnitude with opposite sign
- A generalization of the Wald statistic is used to test the joint null hypothesis (2) that neither the Coke nor Pepsi display affects the probability of choosing Coke
- When using maximum likelihood estimators, such as probit and logit, tests based on the likelihood ratio principle are generally preferred
- The idea is much like the F -test
 - One test component is the log-likelihood function value in the unrestricted, full model ($\ln L_U$) evaluated at the maximum likelihood estimates
 - The second ingredient is the log-likelihood function value from the model that is “restricted” by imposing the condition that the null hypothesis is true ($\ln L_R$)
- The restricted probit model is obtained by imposing the condition $\beta_3 = -\beta_4$:

$$\begin{aligned} p_{\text{Coke}} &= E(\text{COKE}) \\ &= \Phi(\beta_1 + \beta_2 \text{PRATIO} + \beta_3 \text{DISP_COKE} + \beta_4 \text{DISP_PEPSI}) \\ &= \Phi(\beta_1 + \beta_2 \text{PRATIO} - \beta_4 \text{DISP_COKE} + \beta_4 \text{DISP_PEPSI}) \\ &= \Phi(\beta_1 + \beta_2 \text{PRATIO} + \beta_4 [\text{DISP_PEPSI} - \text{DISP_COKE}]) \end{aligned}$$

- We have $L_R = -713.6595$
- The likelihood ratio test statistic value is:

$$\begin{aligned} LR &= 2(\ln L_U - \ln L_R) \\ &= 2(-710.9486 - (-713.6595)) \\ &= 5.4218 \end{aligned}$$

- Therefore, since $\chi^2_{(1)} = 3.841 < LR$, we reject the null hypothesis (1).
- To test the null hypothesis (2), use the restricted model

$$E(\text{COKE}) = \Phi(\beta_1 + \beta_2 \text{PRATIO})$$

- In this case, the value of the likelihood ratio test statistic is $LR = 19.55 > \chi^2_{(2)} = 5.991$. Therefore, we reject the null hypothesis that neither the Coke nor Pepsi display has an effect on the choice of Coke.

R Example

```
rm(list=ls())
library(PoEdata)

# Read in the data
data("transport", package="POE5Rdata")
attach(transport)

# Model 2: Logit
logit.mod = glm(auto~dtype, family=binomial(link="logit"),
                data=transport)
summary(logit.mod)

##
## Call:
## glm(formula = auto ~ dtype, family = binomial(link = "logit"),
##      data = transport)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6469  -0.3414  -0.1128   0.3967   2.3012
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.2376     0.7505  -0.317   0.7516
## dtype         0.5311     0.2064   2.573   0.0101 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 29.065  on 20  degrees of freedom
## Residual deviance: 12.332  on 19  degrees of freedom
## AIC: 16.332
##
## Number of Fisher Scoring iterations: 6
```

```

confint(logit.mod)

##                2.5 %    97.5 %
## (Intercept) -1.9602922 1.270941
## dttime      0.2191503 1.078215

# Predict probabilities:
# Case 1: bus takes 30 min longer than car
xdtime <- data.frame(dtime=3) # 30 min difference
predict(logit.mod, xdtime, type = "response", se.fit = TRUE)

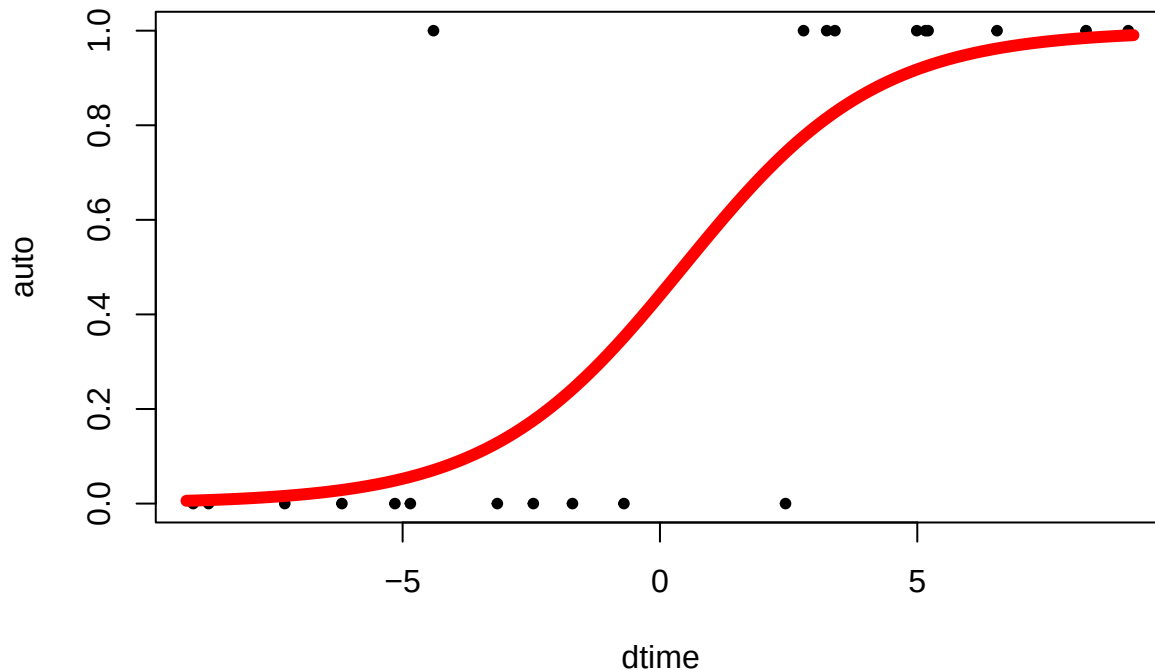
## $fit
##      1
## 0.7950631
##
## $se.fit
##      1
## 0.1451628
##
## $residual.scale
## [1] 1

# Case 2: bus takes 30 min less than car
xdtime <- data.frame(dtime=-3) # 30 min difference
predict(logit.mod, xdtime, type = "response", se.fit = TRUE)

## $fit
##      1
## 0.1381346
##
## $se.fit
##      1
## 0.1248471
##
## $residual.scale
## [1] 1

x = seq(-9.2, 9.2, 0.01)
yhat = predict(logit.mod, list(dtime = x), type = "response", se.fit = TRUE)
plot(dtime, auto, pch=20)
lines(x, yhat$fit, lwd=6, col = "red")

```



4 Multinomial Logit

- We are often faced with choices involving more than two alternatives
- These are called *multinomial choice* situations
 - If you are shopping for a laundry detergent, which one do you choose? Tide, Cheer, Arm & Hammer, Wisk, and so on
 - If you enroll in the business school, will you major in economics, marketing, management, finance, or accounting?
- The estimation and interpretation of the models is, in principle, similar to that in logit and probit models
- The models go under the names:
 - multinomial logit
 - conditional logit
 - multinomial probit
- As in the logit and probit models, we will try to explain the probability that the i^{th} person will choose alternative j

$$p_{ij} = \Pr(\text{Individual } i \text{ chooses alternative } j)$$

- Assume $J = 3$

- For a single explanatory factor, the choice probabilities are:

$$\begin{aligned}
 p_{i1} &= \frac{1}{1 + \exp(\beta_{12} + \beta_{22}x) + \exp(\beta_{13} + \beta_{23}x)}, \\
 p_{i2} &= \frac{\exp(\beta_{12} + \beta_{22}x)}{1 + \exp(\beta_{12} + \beta_{22}x) + \exp(\beta_{13} + \beta_{23}x)}, \\
 p_{i3} &= \frac{\exp(\beta_{13} + \beta_{23}x)}{1 + \exp(\beta_{12} + \beta_{22}x) + \exp(\beta_{13} + \beta_{23}x)},
 \end{aligned} \tag{19}$$

- A distinguishing feature of the multinomial logit model in Eq. 19 is that there is a single explanatory variable that describes the individual, not the alternatives facing the individual
- Such variables are called *individual specific*
- To distinguish the alternatives, we give them different parameter values
- Suppose that we observe three individuals, who choose alternatives 1, 2, and 3, respectively

Assuming that their choices are independent, then the probability of observing this outcome is:

$$\Pr(y_{11} = 1, y_{22} = 1, y_{33} = 1) = p_{11} \times p_{22} \times p_{33}$$

- Or

$$\begin{aligned}
 &\Pr(y_{11} = 1, y_{22} = 1, y_{33} = 1) \\
 &= \frac{1}{1 + \exp(\beta_{12} + \beta_{22}x) + \exp(\beta_{13} + \beta_{23}x)} \\
 &\times \frac{\exp(\beta_{12} + \beta_{22}x)}{1 + \exp(\beta_{12} + \beta_{22}x) + \exp(\beta_{13} + \beta_{23}x)} \\
 &\times \frac{\exp(\beta_{13} + \beta_{23}x)}{1 + \exp(\beta_{12} + \beta_{22}x) + \exp(\beta_{13} + \beta_{23}x)} \\
 &= L(\beta_{12}, \beta_{22}, \beta_{13}, \beta_{23})
 \end{aligned}$$

- Maximum likelihood estimation seeks those values of the parameters that maximize the likelihood or, more specifically, the *log-likelihood function*, which is easier to work with mathematically
- For the value of the explanatory variable x_0 , we can calculate the predicted probabilities of each outcome being selected
- For alternative 1:

$$\hat{p}_{11} = \frac{1}{1 + \exp(\hat{\beta}_{12} + \hat{\beta}_{22}x) + \exp(\hat{\beta}_{13} + \hat{\beta}_{23}x)}$$

- Similarly for alternatives 2 and 3
- The β 's are not "slopes"
- The *marginal effect* is the effect of a change in x , everything else held constant, on the probability that an individual chooses alternative $m = 1, 2$, or 3:

$$\left. \frac{\Delta p_{im}}{\Delta x_i} \right|_{\text{All else constant}} = \frac{\partial p_{im}}{\partial x_i} = p_{im} \left[\beta_{2m} - \sum_{j=1}^3 \beta_{2j} p_{ij} \right] \tag{20}$$

- Alternatively, and somewhat more simply, the difference in probabilities can be calculated for two specific values of x_i

$$\begin{aligned}\hat{\Delta}p_1 &= \hat{p}_{b1} - \hat{p}_{a1} \\ &= \frac{1}{1 + \exp(\hat{\beta}_{12} + \hat{\beta}_{22}x_b) + \exp(\hat{\beta}_{13} + \hat{\beta}_{23}x_b)} \\ &\quad - \frac{1}{1 + \exp(\hat{\beta}_{12} + \hat{\beta}_{22}x_a) + \exp(\hat{\beta}_{13} + \hat{\beta}_{23}x_a)}\end{aligned}$$

- Another useful interpretive device is the *probability ratio*
- It shows how many times more likely category j is to be chosen relative to the first category

$$\frac{\Pr(y_i = j)}{\Pr(y_i = 1)} = \frac{p_{ij}}{p_{i1}} = \exp(\beta_{1j} + \beta_{2j}x), \quad j = 2, 3 \quad (21)$$

- The effect on the probability ratio of changing the value of x_i is given by the derivative:

$$\frac{\partial \left(\frac{p_{ij}}{p_{i1}} \right)}{\partial x_i} = \beta_{2j} \exp(\beta_{1j} + \beta_{2j}x), \quad j = 2, 3 \quad (22)$$

- An interesting feature of the probability ratio Eq. 21 is that it does not depend on how many alternatives there are in total
- There is the implicit assumption in logit models that the probability ratio between any pair of alternatives is *independent of irrelevant alternatives* (IIA):
 - This is a strong assumption, and if it is violated, multinomial logit may not be a good modeling choice
 - It is especially likely to fail if several alternatives are similar
- Tests for the IIA assumption work by dropping one or more of the available options from the choice set and then re-estimating the multinomial model:
 - If the IIA assumption holds, then the estimates should not change very much
 - A statistical comparison of the two sets of estimates, one set from the model with a full set of alternatives, and the other from the model using a reduced set of alternatives, is carried out using a Hausman contrast test proposed by Hausman and McFadden

R Example

```
rm(list=ls())
library(POE5Rdata)
library(caret)
library(nnet)
# Model: Multinomial Logit
data("nels_small", package="POE5Rdata")
attach(nels_small)
```

```
# y = 1 (no college), 2 (two yr), 3 (four yr), x = grades
nels.multinom <- multinom(psechoice~grades, data=nels_small)
```

```
## # weights:  9 (4 variable)
## initial value 1098.612289
## iter  10 value 875.313116
## final value 875.313099
## converged
```

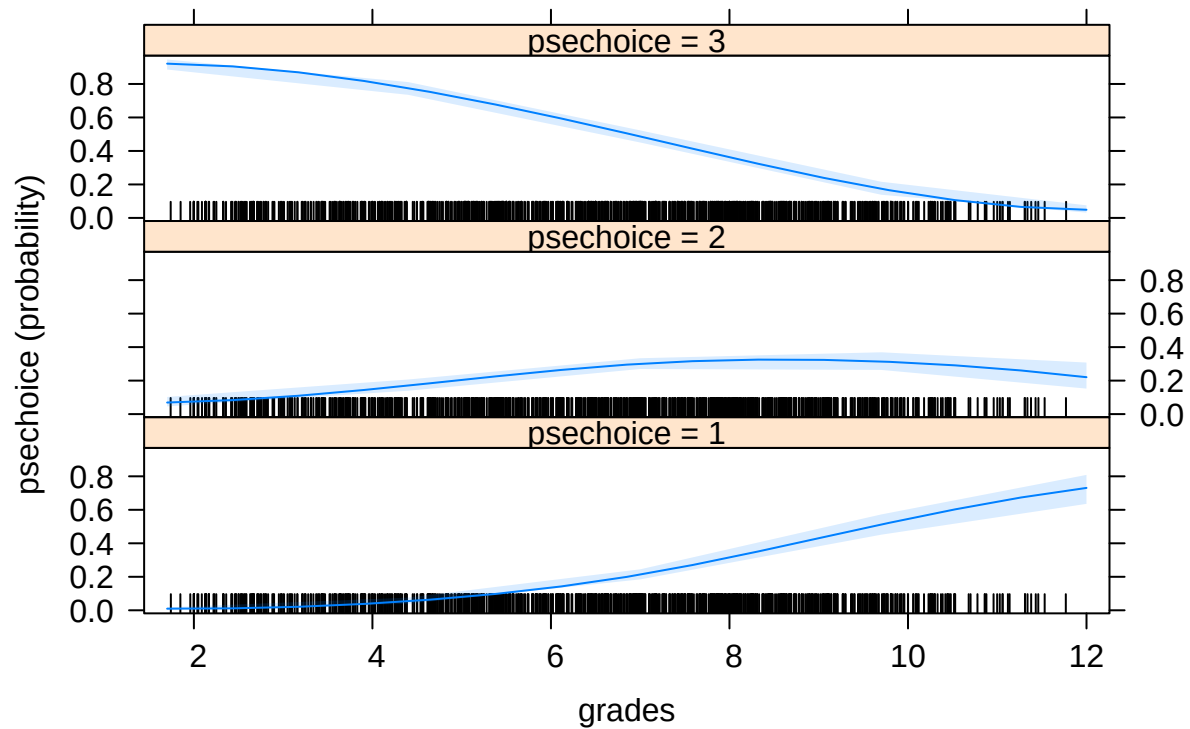
```
summary(nels.multinom)
```

```
## Call:
## multinom(formula = psechoice ~ grades, data = nels_small)
##
## Coefficients:
##      (Intercept)      grades
## 2      2.505273 -0.3086404
## 3      5.770170 -0.7062468
##
## Std. Errors:
##      (Intercept)      grades
## 2      0.4183944 0.05228532
## 3      0.4043290 0.05292638
##
## Residual Deviance: 1750.626
## AIC: 1758.626
```

```
# y = 1 is missing, therefore it is the baseline
```

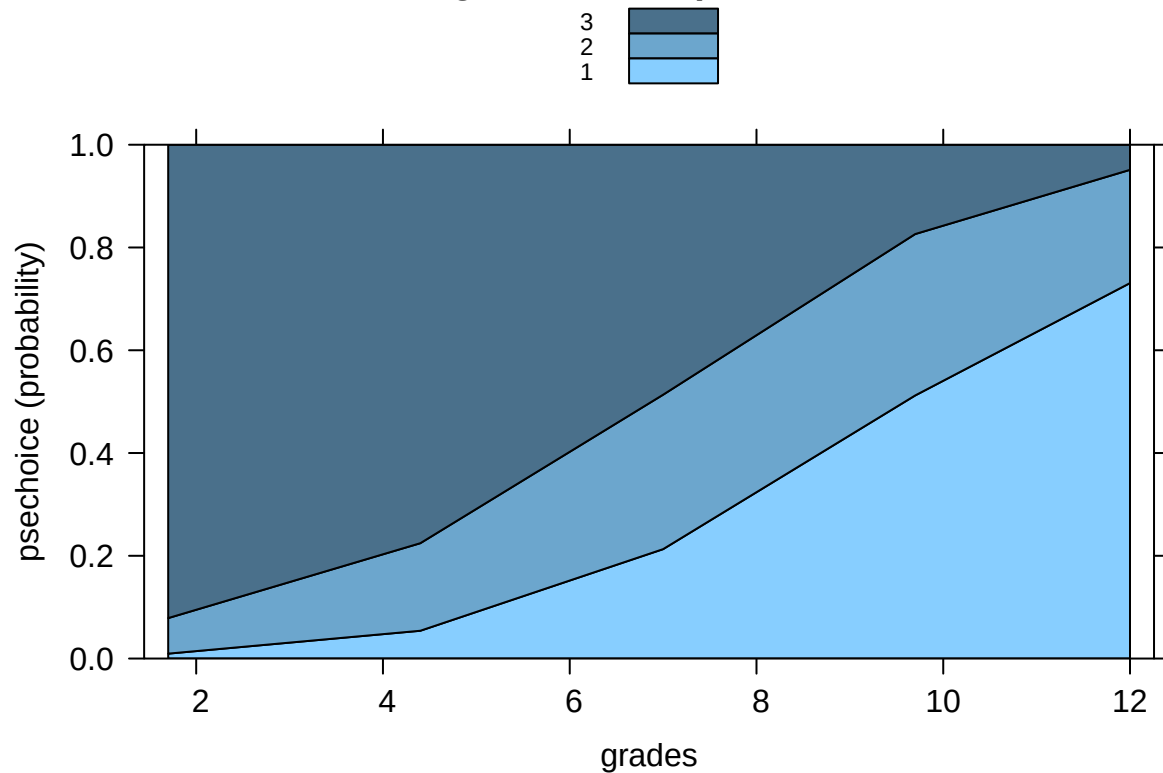
```
# Visualize the probabilities
library("effects")
nels.all <- allEffects(nels.multinom)
plot(nels.all)
```

grades effect plot



```
# Compare all 3 in one plot
plot(nels.all, style = "stacked", colors = c("skyblue1", "skyblue3", "skyblue4"), rug = FALSE)
```

grades effect plot



4.1 Conditional Logit

- Variables like PRICE are *individual-* and *alternative-specific* because they vary from individual to individual and are different for each choice the consumer might make
- This type of information is very different from what we assumed was available in the multinomial logit model, where the explanatory variable x_i was *individual-specific*; it did not change across alternatives
- Consider a model for the probability that individual i chooses alternative j :

$$p_{ij} = \Pr(\text{Individual } i \text{ choose alternative } j)$$

- The conditional logit model specifies these probabilities as:

$$p_{ij} = \frac{\exp(\beta_{1j} + \beta_2 PR_{ij})}{\exp(\beta_{11} + \beta_2 PR_{11}) + \exp(\beta_{12} + \beta_2 PR_{12}) + \exp(\beta_{13} + \beta_2 PR_{13})} \quad (23)$$

where $PR = PRICE$

- Normalization: Set $\beta_{13} = 0$
- Estimation of the unknown parameters is by maximum likelihood
- Suppose that we observe three individuals, who choose alternatives one, two, and three, respectively
- We have:

$$\begin{aligned} & \Pr(y_{11} = 1, y_{22} = 1, y_{33} = 1) \\ &= p_{11} p_{22} p_{33} \\ &= \frac{\exp(\beta_{11} + \beta_2 PR_{11})}{\exp(\beta_{11} + \beta_2 PR_{11}) + \exp(\beta_{12} + \beta_2 PR_{12}) + \exp(\beta_{13} + \beta_2 PR_{13})} \\ &\times \frac{\exp(\beta_{12} + \beta_2 PR_{22})}{\exp(\beta_{11} + \beta_2 PR_{21}) + \exp(\beta_{12} + \beta_2 PR_{22}) + \exp(\beta_{13} + \beta_2 PR_{23})} \\ &\times \frac{\exp(\beta_{13} + \beta_2 PR_{33})}{\exp(\beta_{11} + \beta_2 PR_{31}) + \exp(\beta_{12} + \beta_2 PR_{32}) + \exp(\beta_{13} + \beta_2 PR_{33})} \\ &= L(\beta_{11}, \beta_{12}, \beta_2) \end{aligned}$$

4.2 Conditional Logit Postestimation Analysis

- The own price effect is:

$$\frac{\partial p_{ij}}{\partial PRICE_{ij}} = p_{ij} (1 - p_{ij}) \beta_2 \quad (24)$$

- The change in probability of alternative j being selected if the price of alternative k changes ($k \neq j$) is:

$$\frac{\partial p_{ij}}{\partial PRICE_{ik}} = -p_{ij} p_{ik} \beta_2 \quad (25)$$

- An important feature of the conditional logit model is that the probability ratio between alternatives j and k is:

$$\begin{aligned} \frac{p_{ij}}{p_{ik}} &= \frac{\exp(\beta_{1j} + \beta_2 PRICE_{ij})}{\exp(\beta_{1k} + \beta_2 PRICE_{ik})} \\ &= \exp\{(\beta_{1j} - \beta_{1k}) + \beta_2 (PRICE_{ij} - PRICE_{ik})\} \end{aligned}$$

- The probability ratio depends on the difference in prices, but not on the prices themselves
- Models that do not require the IIA assumption have been developed, but they are difficult
- These include the *multinomial probit* model, which is based on the normal distribution, and the *nested logit* and *mixed logit* models

Example: Coke vs. Pepsi

- The predicted probability of a Pepsi purchase, given that the price of Pepsi is \$1.00, the price of 7-Up is \$1.25 and the price of Coke is \$1.10 is:

$$\hat{p}_{i1} = \frac{\exp(\hat{\beta}_{11} + \hat{\beta}_2 \cdot 1.0)}{\exp(\hat{\beta}_{11} + \hat{\beta}_2 \cdot 1.0) + \exp(\hat{\beta}_{12} + \hat{\beta}_2 \cdot 1.25) + \exp(\hat{\beta}_2 \cdot 1.10)} = 0.4832$$

- The standard error for this predicted probability is 0.0154, which is computed via “the delta method.”
- If we raise the price of Pepsi to \$1.10, we estimate that the probability of its purchase falls to 0.4263 (*se* = 0.0135)
- If the price of Pepsi stays at \$1.00 but we increase the price of Coke by 15 cents, then we estimate that the probability of a consumer selecting Pepsi rises by 0.0445 (*se* = 0.0033)
- These numbers indicate to us the responsiveness of brand choice to changes in prices, much like elasticities

R Example

```
rm(list=ls())
library(POE5Rdata)
library(mclogit)
# Model: Conditional Logit
data(Transport)
# Variables:
# transport = means of transportation that can be chosen --> j=3 (bus, car, train)
# suburb = identifying number for each suburb --> i=10
# distance = walking distance to bus or train station
# cost = cost of each means of transportation
# working = size of working population of each suburb
# prop.true = true choice probabilities
# resp = choice frequencies of means of transportation

clog.mod = mclogit(cbind(resp,suburb)~distance+cost,data=Transport)

##
## Iteration 1 - deviance = 39.74973 - criterion = 0.8590917
## Iteration 2 - deviance = 10.50328 - criterion = 2.758244
## Iteration 3 - deviance = 9.231325 - criterion = 0.1363107
## Iteration 4 - deviance = 9.227742 - criterion = 0.0003840654
## Iteration 5 - deviance = 9.227742 - criterion = 3.446472e-09
```

```
## converged
summary(clog.mod)

##
## Call:
## mclogit(formula = cbind(resp, suburb) ~ distance + cost, data = Transport)
##
##           Estimate Std. Error z value Pr(>|z|)
## distance -1.43940    0.05318  -27.07  <2e-16 ***
## cost      -0.97753    0.03987  -24.52  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Null Deviance:      2734
## Residual Deviance:  9.228
## Number of Fisher Scoring iterations:  5
## Number of observations: 1994

# Note: the deviance is a measure of how far our model is
# from a 'perfect model' and is based on the likelihood ratio
# For details of the variables and mclogit see:
# https://cran.r-project.org/web/packages/mclogit/mclogit.pdf
```

5 Ordered Choice Models

- The choice options in multinomial and conditional logit models have no natural ordering or arrangement
- However, in some cases choices are ordered in a specific way
- Examples:
 1. Results of opinion surveys in which responses can be strongly disagree, disagree, neutral, agree or strongly agree
 2. Assignment of grades or work performance ratings
 3. Standard and Poor's rates bonds as *AAA*, *AA*, *A*, *BBB*, and so on
 4. Levels of employment are unemployed, part-time, or full-time
- When modeling these types of outcomes numerical values are assigned to the outcomes, but the numerical values are *ordinal*, and reflect only the ranking of the outcomes
- In the first example, we might assign a dependent variable y the values:

$$y = \begin{cases} 1 & \text{strongly disagree} \\ 2 & \text{disagree} \\ 3 & \text{neutral} \\ 4 & \text{agree} \\ 5 & \text{strongly agree} \end{cases}$$

- There may be a natural ordering to college choice
- We might rank the possibilities as:

$$y = \begin{cases} 3 & \text{4-year college} \\ 2 & \text{2-year college} \\ 1 & \text{no college} \end{cases} \quad (26)$$

- The usual linear regression model is not appropriate for such data, because in regression we would treat the y values as having some numerical meaning when they do not
- When faced with a ranking problem, we develop a “sentiment” about how we feel concerning the alternative choices, and the higher the sentiment, the more likely a higher-ranked alternative will be chosen
- This sentiment is, of course, unobservable to the econometrician
- Unobservable variables that enter decisions are called *latent variables*

Example: College choice.

- For college choice, a latent variable may be grades:

$$y_i^* = \beta \cdot \text{GRADES}_i + e_i$$

- This model is not a regression model, because the dependent variable is unobservable
- Consequently it is sometimes called an *index model*
- We can now specify:

$$y = \begin{cases} 3 \text{ (4-year college)} & \text{if } y_i^* > \mu_2 \\ 2 \text{ (2-year college)} & \text{if } \mu_1 < y_i^* \leq \mu_2 \\ 1 \text{ (no college)} & \text{if } y_i^* \leq \mu_1 \end{cases} \quad (27)$$

- If we assume that the errors have the standard normal distribution, $N(0,1)$, an assumption that defines the *ordered probit model*, then we can calculate the following:

$$\begin{aligned} \Pr(y_i = 1) &= \Pr(y_i^* \leq \mu_1) = \Pr(\beta \cdot \text{GRADES}_i + e_i \leq \mu_1) \\ &= \Pr(e_i \leq \mu_1 - \beta \cdot \text{GRADES}_i) \\ &= \Phi(\mu_1 - \beta \cdot \text{GRADES}_i) \end{aligned}$$

- Also:

$$\begin{aligned} \Pr(y_i = 2) &= \Pr(\mu_1 < y_i^* \leq \mu_2) \\ &= \Pr(\mu_1 < \beta \cdot \text{GRADES}_i + e_i \leq \mu_2) \\ &= \Pr(\mu_1 - \beta \cdot \text{GRADES}_i < e_i \leq \mu_2 - \beta \cdot \text{GRADES}_i) \\ &= \Phi(\mu_2 - \beta \cdot \text{GRADES}_i) - \Phi(\mu_1 - \beta \cdot \text{GRADES}_i) \end{aligned}$$

- Finally:

$$\begin{aligned} \Pr(y_i = 3) &= \Pr(y_i^* > \mu_2) = \Pr(\beta \cdot \text{GRADES}_i + e_i > \mu_2) \\ &= \Pr(e_i > \mu_2 - \beta \cdot \text{GRADES}_i) \\ &= 1 - \Phi(\mu_2 - \beta \cdot \text{GRADES}_i) \end{aligned}$$

- If we observe a random sample of $N = 3$ individuals, with the first not going to college ($y_1 = 1$), the second attending a two-year college ($y_2 = 2$), and the third attending a four-year college ($y_3 = 3$), then the likelihood function is:

$$L(\beta, \mu_1, \mu_2) = \Pr(y_1 = 1) \Pr(y_2 = 2) \Pr(y_3 = 3)$$

- Econometric software includes options for both *ordered probit*, which depends on the errors being standard normal, and *ordered logit*, which depends on the assumption that the random errors follow a logistic distribution
- Most economists will use the normality assumption
- Many other social scientists use the logistic
- Practically, there is little difference between the results

5.1 The types of questions we can answer with this model

- What is the probability that a high school graduate with $GRADES = 2.5$ (on a 13-point scale, with one being the highest) will attend a two-year college?

$$\Pr(y = 2 | GRADES = 2.5) = \Phi(\hat{\mu}_2 - \hat{\beta} \times 2.5) - \Phi(\hat{\mu}_1 - \hat{\beta} \times 2.5)$$

- What is the difference in probability of attending a two-year college for two students, one with $GRADES = 2.5$ and another with $GRADES = 4.5$?

$$\Pr(y = 2 | GRADES = 4.5) - \Pr(y = 2 | GRADES = 2.5)$$

- If we treat $GRADES$ as a continuous variable, what is the marginal effect on the probability of each outcome, given a one-unit change in $GRADES$?

$$\begin{aligned} \frac{\partial \Pr(y = 1)}{\partial GRADES} &= -\phi(\mu_1 - \beta \cdot GRADES_i) \times \beta \\ \frac{\partial \Pr(y = 2)}{\partial GRADES} &= [\phi(\mu_1 - \beta \cdot GRADES_i) \\ &\quad - \phi(\mu_2 - \beta \cdot GRADES_i)] \times \beta \\ \frac{\partial \Pr(y = 3)}{\partial GRADES} &= \phi(\mu_2 - \beta \cdot GRADES_i) \times \beta \end{aligned}$$

R Example

```
rm(list=ls())
library(POE5Rdata)
library(caret)
library(nnet)
# Model: Ordered Choice
data("nels_small", package="POE5Rdata")
attach(nels_small)
# y = 1 (no college), 2 (two yr), 3 (four yr), x = grades
nels.multinom <- multinom(psechoice~grades, data=nels_small)
```

```
## # weights:  9 (4 variable)
## initial  value 1098.612289
## iter   10 value 875.313116
## final   value 875.313099
## converged
```

```
summary(nels.multinom)
```

```
## Call:
## multinom(formula = psechoice ~ grades, data = nels_small)
##
## Coefficients:
##   (Intercept)    grades
## 2    2.505273 -0.3086404
## 3    5.770170 -0.7062468
##
## Std. Errors:
##   (Intercept)    grades
## 2   0.4183944 0.05228532
## 3   0.4043290 0.05292638
```

```
##
## Residual Deviance: 1750.626
## AIC: 1758.626
# y = 1 is missing, therefore it is the baseline
# For a 1 unit increase in grades (recall that higher is worse),
# the logit coefficient for 2 yr college is expected to decrease
# by 0.3086, for 3 yr college, 0.706

table(nels_small$psechoice)

##
##      1      2      3
## 222 251 527

library(stargazer)
stargazer(nels.multinom, type = "text")

##
## =====
##                               Dependent variable:
##                               -----
##                               2          3
##                               (1)       (2)
## -----
## grades          -0.309***      -0.706***
##                  (0.052)        (0.053)
##
## Constant         2.505***        5.770***
##                  (0.418)        (0.404)
##
## -----
## Akaike Inf. Crit.  1,758.626      1,758.626
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

6 Models for Count Data

- When the dependent variable in a regression model is a count of the number of occurrences of an event, the outcome variable is $y = 0, 1, 2, 3, \dots$
- These numbers are actual counts, and thus different from ordinal numbers
- Examples:
 - The number of trips to a physician a person makes during a year
 - The number of fishing trips taken by a person during the previous year
 - The number of children in a household
 - The number of automobile accidents at a particular intersection during a month
 - The number of televisions in a household
 - The number of alcoholic drinks a college student takes in a week
- The probability distribution we use as a foundation is the Poisson, not the normal or the logistic

- If Y is a Poisson random variable, then its probability function is:

$$f(y) = \Pr(Y = y) = \frac{e^{-\lambda} \lambda^y}{y!}, \quad y = 0, 1, 2, \dots$$

where

$$y! = y \times (y-1) \times (y-2) \times \dots \times 1$$

- In a regression model, we try to explain the behavior of $E(Y)$ as a function of some explanatory variables
- We do the same here, keeping the value of $E(Y) \geq 0$ by defining:

$$E(Y) = \lambda = \exp(\beta_1 + \beta_2 x) \quad (28)$$

- This choice defines the *Poisson regression* model for count data
- Suppose we randomly select $N = 3$ individuals from a population and observe that their counts are $y_1 = 0, y_2 = 2$, and $y_3 = 2$, indicating 0, 2, and 2 occurrences of the event for these three individuals
- The likelihood function is the joint probability function of the observed data is:

$$L(\beta_1, \beta_2) = \Pr(Y = 0) \times \Pr(Y = 2) \times \Pr(Y = 2)$$

- The log-likelihood function is:

$$\ln L(\beta_1, \beta_2) = \ln \Pr(Y = 0) + \ln \Pr(Y = 2) + \ln \Pr(Y = 2)$$

- Using Eq. 28 for λ , the log of the probability function is:

$$\begin{aligned} \ln \Pr(Y = y) &= \ln \left[\frac{e^{-\lambda} \lambda^y}{y!} \right] \\ &= -\lambda + y \ln \lambda - \ln(y!) \\ &= -\exp(\beta_1 + \beta_2 x) + y \times (\beta_1 + \beta_2 x) - \ln(y!) \end{aligned}$$

- Then the log-likelihood function, given a sample of N observations, becomes:

$$\ln L(\beta_1, \beta_2) = \sum_{i=1}^N \{-\exp(\beta_1 + \beta_2 x_i) + y_i \times (\beta_1 + \beta_2 x_i) - \ln(y_i!)\}$$

- Prediction of the conditional mean of y is straightforward:

$$\hat{E}(y_0) = \hat{\lambda}_0 = \exp(\hat{\beta}_1 + \hat{\beta}_2 x_0)$$

- The probability of a particular number of occurrences can be estimated by inserting the estimated conditional mean into the probability function, as:

$$\hat{\Pr}(Y = y) = \frac{\exp(\hat{\lambda}) \hat{\lambda}^y}{y!}, \quad y = 0, 1, 2, \dots$$

- The marginal effect is:

$$\frac{\partial E(y_i)}{\partial x_i} = \lambda_i \beta_2 \quad (29)$$

- This can be expressed as a percentage, which can be useful:

$$\frac{\% \Delta E(y_i)}{\Delta x_i} = 100 \frac{\partial E(y_i) / E(y_i)}{\partial x_i} = 100 \beta_2 \%$$

- Suppose that the conditional mean function contains a indicator variable, how do we calculate its effect?
- If

$$E(y_i) = \lambda_i = \exp(\beta_1 + \beta_2 x_i + \delta D_i),$$

we can examine the conditional expectation when $D = 0$ and when $D = 1$

$$E(y_i|D_i = 0) = \exp(\beta_1 + \beta_2 x_i),$$

$$E(y_i|D_i = 1) = \exp(\beta_1 + \beta_2 x_i + \delta),$$

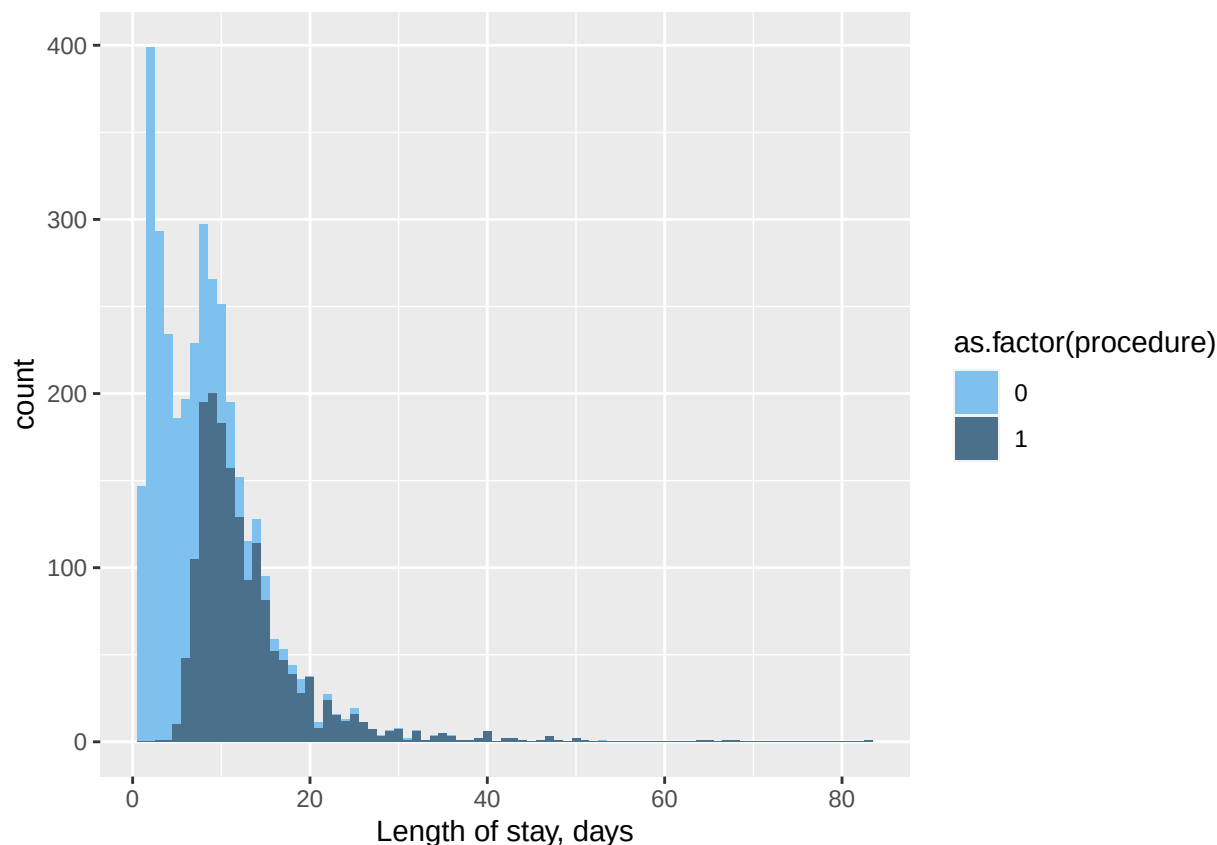
- The percentage change in the conditional mean is:

$$100 \left[\frac{\exp(\beta_1 + \beta_2 x_i + \delta) - \exp(\beta_1 + \beta_2 x_i)}{\exp(\beta_1 + \beta_2 x_i)} \right] \% = 100 [e^\delta - 1] \%$$

R Example

```
rm(list=ls())
library(POE5Rdata)
library(caret)
# Model: Poisson Regression
library(tidyverse)
library(COUNT)
# Model the differential length of stay for patients entering the hospital to receive
# one of two standard cardiovascular procedures: CABG and PTCA
# Data: length of hospital stay (measured in days, y= los), x1= procedure (0=PTCA, 1=CABG),
# x2= admit (0 = elective and 1 = urgent), x3 =sex (male =1, female =0)
data(azpro)

ggplot(data.frame(azpro))+
  geom_histogram(aes(x=los, fill=as.factor(procedure)), binwidth=1)+
  scale_fill_manual(values=c("skyblue2", "skyblue4"))+
  labs(x="Length of stay, days")
```



```
pois.mod = glm(log~ procedure + admit + sex, family=poisson, data=azpro)
summary(pois.mod)
```

```
##
## Call:
## glm(formula = log ~ procedure + admit + sex, family = poisson,
##      data = azpro)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -3.1987  -1.1451  -0.4756   0.5331  12.4784
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.49140    0.01539   96.91  <2e-16 ***
## procedure      0.95738    0.01218   78.61  <2e-16 ***
## admit         0.33307    0.01210   27.52  <2e-16 ***
## sex          -0.13022    0.01179  -11.04  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 16265.0  on 3588  degrees of freedom
## Residual deviance:  8968.9  on 3585  degrees of freedom
## AIC: 22483
##
```

```
## Number of Fisher Scoring iterations: 5
```

```
# Interpretation:  
exp(coef(pois.mod))
```

```
## (Intercept)    procedure      admit      sex  
## 4.4433339    2.6048728    1.3952480    0.8779003
```

```
# procedure: 2.6 days --> The CABG procedure extends the length of stay by ~2.6 days  
# admit: 1.4 days --> An urgent hospital admit extends the length of stay by about 1.4 days  
# sex: 0.9 days --> For males, the the length of stay is shorter by ~ 1 day
```

```
# Get the estimates (in days) w/ conf intervals, and stat. sig:  
tableone::ShowRegTable(pois.mod)
```

```
##          exp(coef) [confint] p  
## (Intercept) 4.44 [4.31, 4.58] <0.001  
## procedure   2.60 [2.54, 2.67] <0.001  
## admit       1.40 [1.36, 1.43] <0.001  
## sex         0.88 [0.86, 0.90] <0.001
```

7 Limited Dependent Variables

- When a model has a discrete dependent variable, the usual regression methods we have studied must be modified
- Now we present another case in which standard least squares estimation of a regression model fails
- This is an example of *censored data*, meaning that a substantial fraction of the observations on the dependent variable take a limit value
- We previously showed the probability density functions for the dependent variable y , at different x -values, centered on the regression function

$$E(y|x) = \beta_1 + \beta_2 x \quad (30)$$

- This leads to sample data being scattered along the regression function
- Least squares regression works by fitting a line through the center of a data scatter, and in this case such a strategy works fine, because the true regression function also fits through the middle of the data scatter
- For our new problem when a substantial number of observations have dependent variable values taking the limit value of zero, the regression function $E(y|x)$ is no longer given by Eq. 30
- Instead $E(y|x)$ is a complicated nonlinear function of the regression parameters β_1 and β_2 , the error variance σ^2 , and x
- The least squares estimators of the regression parameters obtained by running a regression of y on x are biased and inconsistent—least squares estimation fails

Monte Carlo Experiment Example

The data are a simulated sample (*tobit.dat*).

- In this example we give the parameters the specific values $\beta_1 = -9$ and $\beta_2 = 1$
- The observed sample is obtained within the framework of an index or latent variable model:

$$y_i^* = \beta_1 + \beta_2 x_i + e_i = -9 + x_i + e_i \quad (31)$$

- We assume:

$$e_i \sim N(0, \sigma^2 = 16)$$

$$y_i = \begin{cases} 0 & \text{if } y_i^* \leq 0 \\ y_i^* & \text{if } y_i^* > 0 \end{cases}$$

- The estimated regression function for the 200 observed y -values, is given by:

$$\begin{array}{cc} \hat{y} = & -2.1477 & +0.5161x \\ (se) & (0.3706) & (0.0326) \end{array} \quad (32)$$

- If we restrict our sample to include only the 100 positive y -values, the fitted regression is:

$$\begin{array}{cc} \hat{y} = & -3.1399 & +0.6388x \\ (se) & (1.2055) & (0.0827) \end{array} \quad (33)$$

- We can compute the average values of the estimates, which is the Monte Carlo “expected value”:

$$E_{MC}(b_k) = \frac{1}{NSAM} \sum_{m=1}^{NSAM} b_{k(m)} \quad (34)$$

where $\sum_{m=1}^{NSAM} b_{k(m)}$ is the estimate of β_k in the m^{th} Monte Carlo sample

- If the dependent variable is censored, having a lower limit and/or an upper limit, then the least squares estimators of the regression parameters are biased and inconsistent
- We can apply an alternative estimation procedure, which is called *Tobit*

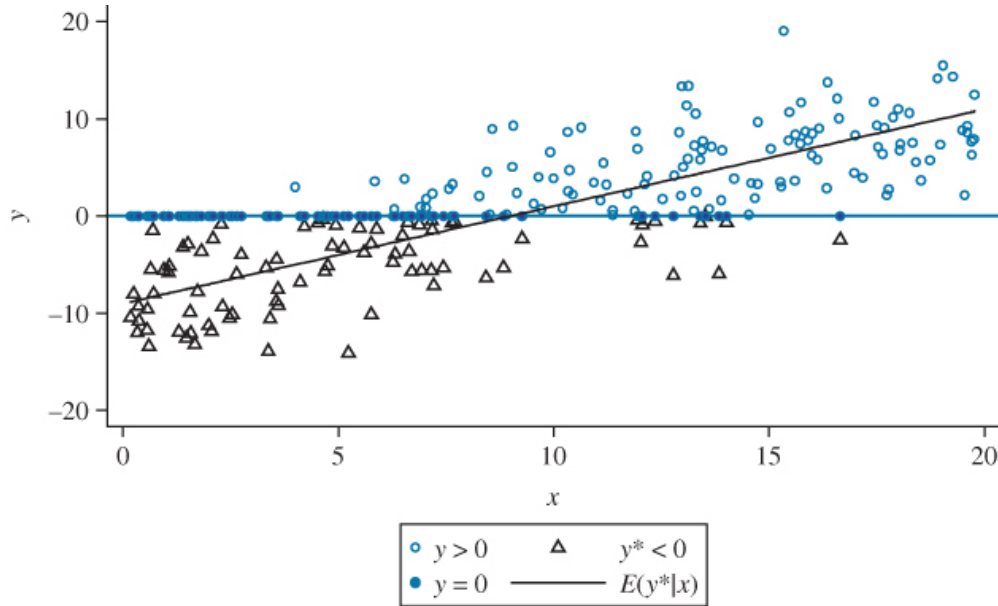
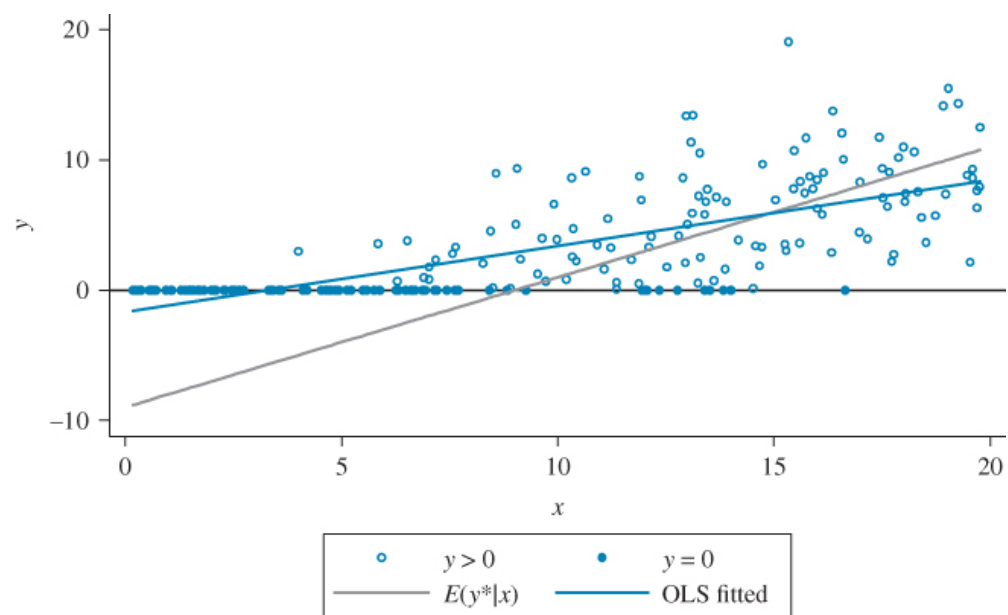


Figure 3: Uncensored sample data and regression function.



7.1 Tobit Model

- *Tobit* is a maximum likelihood procedure that recognizes that we have data of two sorts:
 1. The limit observations ($y = 0$)
 2. The nonlimit observations ($y > 0$)
- The two types of observations that we observe, the limit observations and those that are positive, are generated by the latent variable y^* crossing the zero threshold or not crossing that threshold
- The (probit) probability that $y = 0$ is:

$$\Pr(y_i = 0) = \Pr(y_i^* \leq 0) = 1 - \Phi\left(\frac{\beta_1 + \beta_2 x_i}{\sigma}\right)$$

- The full likelihood function is the product of the probabilities that the limit observations occur times the probability density functions for all the positive, nonlimit, observations:

$$\begin{aligned} L(\beta_1, \beta_2, \sigma^2) &= \prod_{y_i=0} \left(1 - \Phi\left(\frac{\beta_1 + \beta_2 x_i}{\sigma}\right)\right) \\ &\quad \times \prod_{y_i>0} \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} (y_i - \beta_1 - \beta_2 x_i)^2\right\}\right) \end{aligned}$$

- The maximum likelihood estimator is consistent and asymptotically normal, with a known covariance matrix.
- For the Monte Carlo simulated data, we estimate:

$$\begin{array}{cc} \hat{y}_i = & -10.2773 \quad 1.0487x_i \\ (se) & (1.0970) \quad (0.0790) \end{array} \quad (35)$$

- In the Tobit model the parameters β_1 and β_2 are the intercept and slope of the latent variable
- In practice we are interested in the marginal effect of a change in x on either the regression function of the observed data $E(y|x)$ or the regression function conditional on $y > 0$, $E(y|x, y > 0)$
- The slope of $E(y|x)$ is:

$$\frac{\partial E(y|x)}{\partial x} = \beta_2 \Phi\left(\frac{\beta_1 + \beta_2 x}{\sigma}\right) \quad (36)$$

- The marginal effect can be decomposed into two factors called the “McDonald-Moffit” decomposition:

$$\frac{\partial E(y|x)}{\partial x} = \Pr(y > 0) \frac{\partial E(y|x, y > 0)}{\partial x} + E(y|x, y > 0) \frac{\partial \Pr(y > 0)}{\partial x}$$

- The first factor accounts for the marginal effect of a change in x for the portion of the population whose y -data is observed already
- The second factor accounts for changes in the proportion of the population who switch from the y -unobserved category to the y -observed category when x changes

Example: Wage Equation.

- Consider the regression model:

$$HOURS = \beta_1 + \beta_2 EDUC + \beta_3 EXPER + \beta_4 AGE + \beta_5 KIDSL6 + e \quad (37)$$

- The calculated scale factor is $\hat{\Phi} = 0.3638$

- The marginal effect on observed hours of work of another year of education is:

$$\frac{\partial E(HOURS)}{\partial EDUC} = \hat{\beta}_2 \hat{\Phi} = 73.29 \times 0.3638 = 26.34$$

- Another year of education is expected to increase a wife's hours of work by about 26 hours, conditional upon the assumed values of the explanatory variables
- If the data are obtained by random sampling, then classic regression methods, such as least squares, work well
- However, if the data are obtained by a sampling procedure that is not random, then standard procedures do not work well
- Economists regularly face such data problems
- If we wish to study the determinants of the wages of married women, we face a *sample selection* problem
- We only observe data on market wages when the woman chooses to enter the workforce
- If we observe only the working women, then our sample is not a random sample:
The data we observe are "selected" by a systematic process for which we do not account
- A solution to this problem is a technique called *Heckit*
- This procedure uses two estimation steps:

Step 1 A probit model is first estimated explaining why a woman is in the labor force or not. This is the selection equation

Step 2 A least squares regression is estimated relating the wage of a working woman to education, experience, and so on, and a variable called the "inverse Mills ratio," or IMR

The IMR is created from the first step probit estimation and accounts for the fact that the observed sample of working women is not random

7.2 The Selection Equation

- It is expressed in terms of a latent variable z_i^* that depends on one or more explanatory variables w_i , and is given by:

$$z_i^* = \gamma_1 + \gamma_2 w_i + u_i, \quad i = 1, \dots, N \quad (16.37)$$

- The latent variable is not observed, but we do observe the indicator variable:

$$z_i = \begin{cases} 1 & z_i^* > 0 \\ 0 & \text{otherwise} \end{cases} \quad (38)$$

- The second equation is the linear model of interest:

$$y_i = \beta_1 + \beta_2 x_i + e_i, \quad i = 1, \dots, n; \quad \text{with } N > n \quad (39)$$

- A *selectivity problem* arises when y_i is observed only when $z_i = 1$ and if the errors of the two equations are correlated
- In such a situation the usual least squares estimators of β_1 and β_2 are biased and inconsistent

- Consistent estimators are based on the conditional regression function:

$$E(y_i | z_i^* > 0) = \beta_1 + \beta_2 x_i + \beta_\lambda \lambda_i, \quad i = 1, \dots, n \quad (40)$$

where the additional variable λ_i is the “inverse Mills ratio,” namely:

$$\lambda_i = \frac{\phi(\gamma_1 + \gamma_2 w_i)}{\Phi(\gamma_1 + \gamma_2 w_i)} \quad (41)$$

- The parameters γ_1 and γ_2 can be estimated using a probit model, based on the observed binary outcome z_i , so that the estimated IMR:

$$\hat{\lambda}_i = \frac{\phi(\hat{\gamma}_1 + \hat{\gamma}_2 w_i)}{\Phi(\hat{\gamma}_1 + \hat{\gamma}_2 w_i)} \quad (42)$$

- Therefore:

$$y_i = \beta_1 + \beta_2 x_i + \beta_\lambda \hat{\lambda}_i + v_i, \quad i = 1, \dots, n \quad (43)$$

- The estimated model is:

$$\begin{aligned} \hat{\ln}(WAGE) = & -0.4002 + 0.1095 EDUC \\ (t) & (-2.10) \quad (7.73) \\ & + 0.0157 EXPER \quad R^2 = 0.1484 \\ & (3.90) \end{aligned} \quad (44)$$

- The Heckit procedure starts by estimating a probit model:

$$\hat{\Pr}(LFP = 1) = \Phi(1.1923 - 0.0206 AGE + 0.0838 EDUC - 0.3139 MTR) \\ (t) \quad (-2.93) \quad (3.61) \quad (-2.54) \quad (-2.26)$$

- The inverse Mills ratio is:

$$\begin{aligned} \hat{\lambda} &= \frac{\widehat{IMR}}{\Phi(1.1923 - 0.0206 AGE + 0.0838 EDUC - 0.3139 MTR)} \\ &= \frac{\phi(1.1923 - 0.0206 AGE + 0.0838 EDUC - 0.3139 MTR)}{\Phi(1.1923 - 0.0206 AGE + 0.0838 EDUC - 0.3139 MTR)} \end{aligned}$$

- The final combined model is:

$$\begin{aligned} \hat{\ln}(WAGE) = & 0.8105 + 0.0585 EDUC \\ (t) & (1.64) \quad (2.45) \\ (t\text{-adjusted}) & (1.33) \quad (1.97) \\ & + 0.0163 EXPER \quad -0.8664 IMR \\ & (4.08) \quad (-2.65) \\ & (3.88) \quad (-2.17) \end{aligned} \quad (45)$$

- In most instances it is preferable to estimate the full model, both the selection equation and the equation of interest, jointly by maximum likelihood
- The maximum likelihood estimated wage equation is:

$$\hat{\ln}(WAGE) = 0.6686 + 0.0658 EDUC + 0.0118 EXPER \\ (t) \quad (2.84) \quad (3.96) \quad (2.87)$$

- The standard errors based on the full information maximum likelihood procedure are smaller than those yielded by the two-step estimation method

R Example

```
rm(list=ls())
# Model: Tobit
library(POE5Rdata)
library(AER)
data(mroz)
tobit.mod <- tobit(hours~educ+exper+age+kidsl6, data=mroz)
summary(tobit.mod)
```

```
##
## Call:
## tobit(formula = hours ~ educ + exper + age + kidsl6, data = mroz)
##
## Observations:
##           Total   Left-censored   Uncensored Right-censored
##           753         325         428             0
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) 1349.8763   386.2991   3.494 0.000475 ***
## educ         73.2910    20.4746   3.580 0.000344 ***
## exper        80.5353     6.2878  12.808 < 2e-16 ***
## age        -60.7678     6.8882  -8.822 < 2e-16 ***
## kidsl6     -918.9181    111.6607  -8.230 < 2e-16 ***
## Log(scale)    7.0332     0.0371 189.565 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Scale: 1134
##
## Gaussian distribution
## Number of Newton-Raphson Iterations: 4
## Log-likelihood: -3827 on 6 Df
## Wald-statistic: 243.3 on 4 Df, p-value: < 2.22e-16
```

```
# We can also use the Censored Regression library
library(censReg)
tobit.cr <- censReg(hours~educ+exper+age+kidsl6, data=mroz)
summary(tobit.cr)
```

```
##
## Call:
## censReg(formula = hours ~ educ + exper + age + kidsl6, data = mroz)
##
## Observations:
##           Total   Left-censored   Uncensored Right-censored
##           753         325         428             0
##
## Coefficients:
##           Estimate Std. error t value Pr(> t)
## (Intercept) 1349.8763   386.2993   3.494 0.000475 ***
## educ         73.2910    20.4746   3.580 0.000344 ***
## exper        80.5353     6.2878  12.808 < 2e-16 ***
## age        -60.7678     6.8882  -8.822 < 2e-16 ***
## kidsl6     -918.9181    111.6607  -8.230 < 2e-16 ***
```

```

## logSigma      7.0332      0.0371 189.565 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Newton-Raphson maximisation, 9 iterations
## Return code 1: gradient close to zero (gradtol)
## Log-likelihood: -3827.143 on 6 Df
margEff( tobit.cr, xValues = c(740.57,12.29, 10.63, 42.54, 1))

##      educ      exper      age      kidsl6
## 73.29099  80.53527 -60.76780 -918.91808

# Phi = 0.3630
# Multiply the MEs by Phi to the answer in "hours worked"
# educ = 73.29 --> ME = 73.29x0.360 = 27

# Sample Selection
library(sampleSelection)
heckit.mod <- selection(lfp~age+educ+I(kids618+kidsl6)+mtr,
                        log(wage)~educ+exper,
                        data=mroz, method="ml")
summary(heckit.mod)

## -----
## Tobit 2 model (sample selection model)
## Maximum Likelihood estimation
## Newton-Raphson maximisation, 4 iterations
## Return code 8: successive function values within relative tolerance limit (reltol)
## Log-Likelihood: -913.5131
## 753 observations (325 censored and 428 observed)
## 10 free parameters (df = 743)
## Probit selection equation:
##      Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.53798    0.61889   2.485  0.0132 *
## age              -0.01346    0.00603  -2.232  0.0259 *
## educ              0.06278    0.02180   2.879  0.0041 **
## I(kids618 + kidsl6) -0.05108    0.03276  -1.559  0.1194
## mtr              -2.20864    0.54620  -4.044 5.81e-05 ***
## Outcome equation:
##      Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.646221    0.235569   2.743  0.00623 **
## educ         0.066462    0.016573   4.010 6.68e-05 ***
## exper        0.011972    0.004085   2.931  0.00348 **
## Error terms:
##      Estimate Std. Error t value Pr(>|t|)
## sigma  0.84112    0.04302   19.55 <2e-16 ***
## rho    -0.82768    0.03911  -21.16 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## -----

```